

Recall from previously:

- If A is an $n \times n$ matrix, then $\lambda \in \mathbb{R}$ is an **eigenvalue** of A if there exists a non-zero vector v , called an **eigenvector** corresponding to λ , such that

$$Av = \lambda v.$$

- If λ is an eigenvalue of an $n \times n$ matrix A , then the set of all solutions v to $Av = \lambda v$ is the **eigenspace** of A corresponding to λ .
- **Theorem 1:** The eigenvalues of a triangular matrix are the diagonal entries.
- **Theorem 2:** If $v_1, \dots, v_p$ are eigenvectors corresponding to distinct eigenvalues $\lambda_1, \dots, \lambda_p$ of an $n \times n$ matrix A , then $v_1, \dots, v_p$ are linearly independent.

- If A is an $n \times n$ matrix, then the characteristic polynomial of A , denoted by $p_A(\lambda)$, is given by

$$p_A(\lambda) = \det(A - \lambda I_n).$$

- λ is an eigenvalue of an $n \times n$ matrix A if and only if λ is a zero of the characteristic polynomial.
- If A and B are similar matrices, then they have the same characteristic polynomials and hence the same eigenvalues (with the same algebraic multiplicities).
- An $n \times n$ matrix A is **diagonalizable** if A is similar to a diagonal matrix.
- **Diagonalization Theorem:** An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors. Moreover, $A = PDP^{-1}$ with D diagonal if and only if the columns of P are n linearly independent eigenvectors of A . In this case the diagonal entries of D are the corresponding eigenvalues of A .
- **Theorem 6:** An $n \times n$ matrix A with n distinct eigenvalues is diagonalizable.
- **Theorem 7:** Let A be an $n \times n$ matrix with distinct eigenvalues $\lambda_1, \dots, \lambda_p$.
 1. For $1 \leq k \leq p$, the dimension of the eigenspace of λ_k is at most the algebraic multiplicity of the eigenvalue λ_k .
 2. The matrix A is diagonalizable if and only if the sum of the dimensions of the eigenspaces is n if and only if the characteristic polynomial factors into linear factors and the dimension for each eigenspace is exactly the algebraic multiplicity.
 3. If A is diagonalizable and $\mathcal{B}_k$ is a basis for the eigenspace corresponding to λ_k , then $\mathcal{B}_1, \dots, \mathcal{B}_p$ is an eigenbasis for $\mathbb{R}^n$.

1. Let $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Find eigenvalues and eigenspaces.

2. Let A be an $n \times n$ matrix with real entries. Suppose λ is an eigenvalue of A , then is $\bar{\lambda}$ an eigenvalue of A .

3. Let $C = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ where a, b are real and not both zero, then find the eigenvalues of C .

Theorem: Let A be a real 2 by 2 matrix with a complex eigenvalue $\lambda = a - bi$ ($b \neq 0$) and an associated eigenvector v in $\mathbb{C}^2$. Then $A = PCP^{-1}$ where $P = [\operatorname{Re}(v) \operatorname{Im}(v)]$ and $C = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$.

4. Prove the theorem.

5. Let $A = \begin{bmatrix} 3 & -2 \\ 4 & -1 \end{bmatrix}$, then find P and C such that $A = PCP^{-1}$ as in the previous theorem.

Complex Eigenvalues— Answers / Solutions

1. We find the eigenvalues using the characteristic polynomial; namely,

$$p_A(\lambda) = \det \begin{bmatrix} -\lambda & -1 \\ 1 & -\lambda \end{bmatrix} = \lambda^2 + 1.$$

Thus the zeros of the characteristic polynomial are $\lambda = \pm i$. We find the eigenspaces now.

$\lambda = i$.

Here we get

$$A - iI_2 = \begin{bmatrix} -i & -1 \\ 1 & -i \end{bmatrix},$$

and hence $E_i = N(A - iI_2) = \text{Span} \left\{ \begin{bmatrix} 1 \\ -i \end{bmatrix} \right\}$.

$\lambda = -i$.

Here we get

$$A + iI_2 = \begin{bmatrix} i & -1 \\ 1 & i \end{bmatrix},$$

and hence $E_{-i} = N(A + iI_2) = \text{Span} \left\{ \begin{bmatrix} 1 \\ i \end{bmatrix} \right\}$.

Thus we can diagonalize A over the complex numbers as

$$A = PDP^{-1} = \begin{bmatrix} 1 & 1 \\ -i & i \end{bmatrix} \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -i & i \end{bmatrix}^{-1}.$$

2. Let A be an $n \times n$ matrix with real entries, then the complex conjugate of A is still just A ; i.e. $\overline{A} = A$. Let λ be an eigenvalue of A with eigenvector v . Then

$$\overline{\lambda v} = \overline{\lambda} \overline{v} = \overline{Av} = \overline{A} \overline{v} = A \overline{v},$$

and hence $\overline{\lambda}$ is also an eigenvalue of A with eigenvector $\overline{v}$.

3. We again use the characteristic polynomial to find the eigenvalues of C ; namely,

$$p_C(\lambda) = \det(C - \lambda I_2) = \det \begin{bmatrix} a - \lambda & -b \\ b & a - \lambda \end{bmatrix} = a^2 + b^2 - 2a\lambda + \lambda^2.$$

Thus using the quadratic formula, we get the zeros of the characteristic polynomial are $\lambda = a \pm bi$.

Expressing $\lambda = a + bi$ in polar form, we get $\lambda = r(\cos \theta + i \sin \theta)$, and hence $a = r \cos \theta$ and $b = r \sin \theta$. Thus

$$C = \begin{bmatrix} a & -b \\ b & a \end{bmatrix} = \begin{bmatrix} r \cos \theta & -r \sin \theta \\ r \sin \theta & r \cos \theta \end{bmatrix} = \begin{bmatrix} r & 0 \\ 0 & r \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}.$$

Thus we can regard multiplying by matrices of the form C as rotating by θ and scaling by r !

4. *Proof.* Let A be a 2×2 matrix with real entries. Suppose A has eigenvalue $\lambda = a - bi$ with $b \neq 0$ and associated complex eigenvector v . Define the change of coordinates matrix P by $P = [\operatorname{Re}(v) \operatorname{Im}(v)]$. We want to prove $A = PCP^{-1}$ where C has the desired form. We will prove that $AP = PC$. We begin by finding $A\operatorname{Re}(v)$ and $A\operatorname{Im}(v)$. Now

$$\begin{aligned} A\operatorname{Re}(v) + iA\operatorname{Im}(v) &= A(\operatorname{Re}(v) + i\operatorname{Im}(v)) \\ &= Av = \lambda v = (a - bi)(\operatorname{Re}(v) + i\operatorname{Im}(v)) \\ &= (a\operatorname{Re}(v) + b\operatorname{Im}(v)) + i(-b\operatorname{Re}(v) + a\operatorname{Im}(v)) \\ &= P \begin{bmatrix} a \\ b \end{bmatrix} + iP \begin{bmatrix} -b \\ a \end{bmatrix}, \end{aligned}$$

and hence $A\operatorname{Re}(v) = P \begin{bmatrix} a \\ b \end{bmatrix}$ and $A\operatorname{Im}(v) = P \begin{bmatrix} -b \\ a \end{bmatrix}$. Thus

$$AP = [A\operatorname{Re}(v) \ A\operatorname{Im}(v)] = \left[P \begin{bmatrix} a \\ b \end{bmatrix} \ P \begin{bmatrix} -b \\ a \end{bmatrix} \right] = P \begin{bmatrix} a & -b \\ b & a \end{bmatrix} = PC.$$

□

5. Finding the eigenvalues and eigenspaces as usual, we find the eigenvalue are $1 \pm 2i$ and the eigenspaces are

$$E_{1+2i} = \operatorname{Span} \left\{ \begin{bmatrix} 1+i \\ 2 \end{bmatrix} \right\},$$

and

$$E_{1-2i} = \operatorname{Span} \left\{ \begin{bmatrix} 1-i \\ 2 \end{bmatrix} \right\}.$$

Using the previous theorem, we take $\lambda = 1 - 2i$ and hence $a = 1$ and $b = 2$ in the Theorem.

Take $P = [\operatorname{Re}(v) \operatorname{Im}(v)] = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}$. Using the previous Theorem, we get

$$A = PCP^{-1} = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}^{-1}.$$